

1. Administrative & Study Identification

**Full Study Title:** Concentration of Ofatumumab in the Breast Milk of Lactating Women With Relapsing Forms of Multiple Sclerosis  
**Official Title:** A Phase IV, Prospective, Multicenter, Open-label, Mother-milk Study to Evaluate Ofatumumab Concentration in the Breast Milk of Lactating Women With Relapsing Forms of Multiple Sclerosis Receiving Ofatumumab  
**Short Title/Acronym:** KATHAROS  
**Protocol Number:** COMB157G2410 **NCT Number:** NCT06444113 **Trial Status:** Recruiting **Sponsor Name:** Novartis Pharmaceuticals  
**Investigational Product:** Ofatumumab (Anti-CD20 monoclonal antibody) | **Trade Names:** Arzerra, Kesimpta **Phase of Development:** Phase IV **Medical Monitor:** Novartis Medical Monitor

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate whether ofatumumab is excreted at quantifiable levels, to determine its maximum concentration (Cmax) in breast milk, and to calculate the Milk/Plasma (M/P) ratio in the mature breast milk of lactating women with Relapsing Multiple Sclerosis (RMS). **Secondary Objectives:** To evaluate other pharmacokinetic (PK) parameters of ofatumumab in breast milk and plasma, estimate the relative infant dose of ofatumumab, and collect safety data to follow up on the overall health and safety of both the breastfeeding mothers receiving ofatumumab and their babies for up to 1 year.

**2.2 Background & Rationale** To address the knowledge gap regarding the use of disease-modifying therapies (DMTs) during lactation. Expectant mothers and women who wish to breastfeed are often advised to halt DMTs (like anti-CD20 therapies), which subsequently increases their risk of severe MS relapses. This post-marketing study generates real-world pharmacokinetic (PK) evidence to determine if it is safe for women to initiate or re-initiate ofatumumab post-partum while continuing to exclusively breastfeed.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Prospective, Observational/Interventional (Phase IV) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

**3.2 Planned Enrollment & Duration Target \$N\$:** 20 participants  
**Number of Sites:** 22 locations (Multicenter) **Estimated Study Start:** 25-Nov-2024 **Estimated Primary Completion:** 23-Feb-2027

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Written informed consent must be obtained before any study assessment is performed. 2. Female participants with a relapsing form of MS (RMS) and at least 18 years of age at the time of providing consent. 3. Must be postpartum at the time of enrollment, delivered a term infant (at least 37 weeks gestation), plan to be exclusively breastfeeding, and willing to provide breast milk samples. 4. Plan to initiate or re-initiate treatment with ofatumumab between 2 to 24 weeks postpartum. The decision to be treated with ofatumumab and to breastfeed is made in accordance with the treating physician and must be completely independent of the decision to participate in this study.

**4.2 Exclusion Criteria** 1. Use of any investigational drugs within 5 half-lives of enrollment, or within 30 days (or until the expected pharmacodynamic effect has returned to baseline), whichever is longer. 2. Participant taking medications prohibited by the study protocol at screening. 3. Pregnant woman, confirmed by a positive serum pregnancy test during screening. 4. Female of childbearing potential not using effective contraception as per local label. 5. History of chronic alcohol abuse or drug abuse in the last year. 6. Any medical, obstetrical, psychiatric, or other condition that, in the opinion of the Investigator, could jeopardize participation or confound study assessments. 7. History of breast implants, breast augmentation, or breast reduction surgery. 8. Received anti-CD20 agents during the second and third trimesters of pregnancy. 9. Active infections (including mastitis; participant may be included once the infection is resolved). 10. Prior or current history of primary or secondary immunodeficiency, or otherwise severely immunocompromised state. 11. Active hepatitis B disease prior to the initiation or re-initiation of ofatumumab. 12. History of malignancy of any organ system (other than localized basal cell carcinoma of the skin or in situ cervical cancer), treated or untreated, within the past 5 years. 13. Any contraindication as per the local label. 14. Participant who has an infant with any abnormality that may interfere with breastfeeding or confound the study assessment.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Standard prescribed ofatumumab maintenance doses initiated or re-initiated 2 to 24 weeks postpartum. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Mothers and babies participate for up to 1 year (up to 12 weeks in the sampling period and approximately 9 months for the safety follow-up part).

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard concomitant therapies for MS management that are not contraindicated during lactation. **Prohibited:** Use of any investigational drugs within 5 half-lives of enrollment, or within 30 days (whichever is longer), and any other medications prohibited by the study protocol.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window / Duration | Key Activities/Procedures                                                                                                                  |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Screening         | Up to 4 weeks   Informed Consent, Eligibility Assessment, Pregnancy Test                                                                   |
| 2        | Sampling Period   | Up to 12 weeks   3 On-site visits (Blood draws pre-dose); 4 Remote visits (Breast milk collection via mobile nursing team, ~6 times total) |
| 3        | Safety Follow-Up  | ~ 9 months   Monitoring health, safety, and development of mothers and their babies                                                        |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Quantification of ofatumumab concentration in the breast milk of lactating women with RMS who have initiated or re-initiated treatment post-partum (including Maximum concentration (C<sub>max</sub>) over 28 days after the second or any subsequent maintenance dose, and the Milk/Plasma (M/P) ratio). **Measurement Method:** PK quantification via paired mature breast milk samples and maternal blood plasma samples.

**7.2 Secondary & Exploratory Endpoints** \* Evaluate other PK parameters of ofatumumab in the breast milk and plasma of lactating women with RMS. \* Estimation of the relative infant dose of ofatumumab. \* Safety data collected in lactating women receiving ofatumumab and their breastfed infants (including long-term health, safety, and developmental follow-up of the infants for up to 1 year).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring during the 1-year study duration, heavily focused on the 9-month Safety Follow-up Part for both mother and infant. **Serious Adverse Events (SAEs):** Standard 24-hour expedited reporting protocols apply. **Data Monitoring Committee (DMC):** Managed by Novartis to continually assess maternal and infant safety.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Pharmacokinetic (PK) analysis set for lactating mothers with quantifiable samples; Safety Set for all enrolled mothers and their babies. **Sample Size Justification:**  $N=20$ , which is sufficiently powered for a specialized, multicenter PK lactation evaluation. **Handling of Missing Data:** Standard PK parameter estimations and safety imputation protocols.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Combined onsite monitoring and remote monitoring facilitated by a mobile nursing team traveling to the patient's location to collect samples. **Audit Trail:** eCRF locking and strict PK sample chain-of-custody tracking.

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## 11. Study Identifiers & Governance

**Primary ID:** COMB157G2410 **Registry Number:** NCT06444113 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** KATHAROS **Official Regulatory Title:** A Phase IV, Prospective, Multicenter, Open-label, Mother-milk Study to Evaluate Ofatumumab Concentration in the Breast Milk of Lactating Women With Relapsing Forms of Multiple Sclerosis Receiving Ofatumumab

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## 12. Clinical Framework (Multiple Sclerosis/Lactation Specific)

**Primary Condition:** Multiple Sclerosis (Relapsing-remitting multiple sclerosis) **Diagnosis Threshold:** Postpartum female (term infant  $\geq 37$  weeks) **Medical Methodology:** Anti-CD20 monoclonal antibody therapy **Optimization Target:** Confirming the Milk/Plasma (M/P) ratio and estimating relative infant dose to support breastfeeding safety guidelines.

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard clinical pathway for postpartum RMS (physician-directed ofatumumab initiation/re-initiation) **Clinical Implementation:** Mobile nursing team dispatch for remote, at-home breast milk collection **Education Protocol (Patient):** Lactation safety monitoring and PK sampling education **Quality Audit Mechanism:** Maternal/infant health tracking and sample viability auditing

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14. Observation & Follow-Up Schedule

**Total Duration:** Up to 1 year **Onsite Visit Cadence:** 3 total onsite clinic visits during sampling **Remote Monitoring Frequency:** 4 remote home visits by mobile nursing team **Checklist Review Interval:** Routine checks continuously throughout the 9-month safety follow-up

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15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |